# Information about Imaginary Numbers, version 0

**Statement:** The imaginary number $i$ is such that $i \neq 0, i \not< 0$, and $i \not> 0$.

**Proof:** Clearly $i \neq 0$ because $i \stackrel{\text{def}}{=} \sqrt{-1}$, so suppose $i < 0$. If we multiply this inequality by $i$ on both sides, we obtain

$$i \cdot i = i^2 = -1 > i \cdot 0 = 0,$$

or simply $-1 > 0$ (note that the direction of the inequality symbol flipped because we supposed $i < 0$). This is a contradiction. Now suppose $i > 0$. Multiplying this inequality by $i$ on both sides yields

$$i \cdot i = i^2 = -1 > i \cdot 0 = 0,$$

i.e., $-1 > 0$, the same contradiction as before. Thus, we conclude that $i \neq 0$, $i \not< 0$, and $i \not> 0$.

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**Euler's Identity ("Euler" is pronounced like "Oiler"):** It can be shown using calculus that

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$
$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$
$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

These are called Maclaurin Series. Let's try adding $\cos(x)$ and $\sin(x)$ together to see what happens. We get

$$\cos(x)+\sin(x) = \left(1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\dots\right)+\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\dots\right)$$
$$= 1+x-\frac{x^2}{2!}-\frac{x^3}{3!}+\frac{x^4}{4!}+\frac{x^5}{5!}-\frac{x^6}{6!}-\frac{x^7}{7!}+\dots$$

Notice that this looks almost like the Maclaurin Series for $e^x$, but some of the signs are off. Can we fix that? Notice that all of the terms with powers that are multiples of 4 have a positive sign in front of them. Furthermore, all the terms with even powers that are not multiples of 4 have negative signs in front of them. This is consistent with how $i$ behaves; for example, $i^2=-1$, $i^4=1$, $i^6=-1$, $i^8=1$ and so on. Therefore, we can write

$$\begin{aligned}
\cos(x)+\sin(x) &= \left(1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\dots\right)+\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\dots\right)\\
&= 1+x-\frac{x^2}{2!}-\frac{x^3}{3!}+\frac{x^4}{4!}+\frac{x^5}{5!}-\frac{x^6}{6!}-\frac{x^7}{7!}+\dots\\
&= 1+x+i^2\frac{x^2}{2!}-\frac{x^3}{3!}+i^4\frac{x^4}{4!}+\frac{x^5}{5!}+i^6\frac{x^6}{6!}-\frac{x^7}{7!}+\dots\\
&= 1+x+\frac{(ix)^2}{2!}-\frac{x^3}{3!}+\frac{(ix)^4}{4!}+\frac{x^5}{5!}+\frac{(ix)^6}{6!}-\frac{x^7}{7!}+\dots
\end{aligned}$$

This makes all of the even terms positive. We still need to make all the terms with odd exponents positive to make this look like the Maclaurin Series for $e^x$. These terms have signs consistent with the odd powers of $i$. For example, $i^1=i$, $i^3=-i$, $i^5=i$, $i^7=-i$, and so on. While the signs are right, there are in fact no $i$'s in these terms. In order to get them in there, we have to actually multiply them by $i$. So let us look at $\cos(x)+i\sin(x)$. We have

$$\begin{aligned}
\cos(x)+i\sin(x) &= 1+ix+\frac{(ix)^2}{2!}-i\frac{x^3}{3!}+\frac{(ix)^4}{4!}+i\frac{x^5}{5!}+\frac{(ix)^6}{6!}-i\frac{x^7}{7!}+\dots\\
&= 1+ix+\frac{(ix)^2}{2!}+\frac{(ix)^3}{3!}+\frac{(ix)^4}{4!}+\frac{(ix)^5}{5!}+\frac{(ix)^6}{6!}+\frac{(ix)^7}{7!}+\dots\\
&= e^{ix}.
\end{aligned}$$

Having established that $e^{ix}=\cos(x)+i\sin(x)$, we can now conclude that

$$e^{i\pi}=\cos(\pi)+i\sin(\pi)=-1+i\cdot 0=-1.$$

In other words,

$$e^{i\pi}+1=0.$$